

Problem

Evaluate the following complex numbers and leave your results in polar form:

(a) $5\angle 30^\circ \left(6 - j8 + \frac{3\angle 60^\circ}{2 + j} \right)$

(b) $\frac{(10\angle 60^\circ)(35\angle -50^\circ)}{(2 + j6) - (5 + j)}$

Step-by-step solution

Step 1 of 5

(a)

Consider the following data:

$$5\angle 30^\circ \left(6 - j8 + \frac{3\angle 60^\circ}{2 + j} \right)$$

Convert the polar number $5\angle 30^\circ$ to complex number.

$$\begin{aligned} 5\angle 30^\circ &= 5[\cos(30^\circ) + j\sin(30^\circ)] \\ &= 5[0.866 + j0.5] \\ &= 4.33 + j2.5 \end{aligned}$$

Convert the polar number $3\angle 60^\circ$ to complex number.

$$\begin{aligned} 3\angle 60^\circ &= 3[\cos(60^\circ) + j\sin(60^\circ)] \\ &= 3[0.5 + j0.866] \\ &= 1.5 + j2.6 \end{aligned}$$

Step 2 of 5

Substitute $1.5 + j2.6$ for $3\angle 60^\circ$ and $4.33 + j2.5$ for $5\angle 30^\circ$ in $5\angle 30^\circ \left(6 - j8 + \frac{3\angle 60^\circ}{2 + j} \right)$.

$$\begin{aligned} 5\angle 30^\circ \left(6 - j8 + \frac{3\angle 60^\circ}{2 + j} \right) &= (4.33 + j2.5) \left(6 - j8 + \frac{1.5 + j2.6}{2 + j} \right) \\ &= (4.33 + j2.5)(6 - j8 + 1.12 + j0.74) \\ &= (4.33 + j2.5)(7.12 - j7.26) \\ &= 48.98 - j13.64 \end{aligned}$$

Step 3 of 5

Convert the result $48.98 - j13.64$ to polar form.

$$\begin{aligned}5\angle 30^\circ \left(6 - j8 + \frac{3\angle 60^\circ}{2 + j} \right) &= 48.98 - j13.64 \\&= \sqrt{48.98^2 + 13.64^2} \angle \tan^{-1} \left(\frac{-13.64}{48.98} \right) \\&= 50.84 \angle -15.56^\circ\end{aligned}$$

Therefore, the polar form of $5\angle 30^\circ \left(6 - j8 + \frac{3\angle 60^\circ}{2 + j} \right)$ is $\boxed{50.84 \angle -15.56^\circ}$.

Step 4 of 5

(b)

Consider the following data:

$$\begin{aligned}\frac{(10\angle 60^\circ)(35\angle -50^\circ)}{(2 + j6) - (5 + j)} &= \frac{(10)(35)\angle -50^\circ + 60^\circ}{(2 + j6 - 5 - j)} \\&= \frac{350\angle 10^\circ}{(-3 + j5)}\end{aligned}$$

Convert $(-3 + j5)$ to polar form.

$$\begin{aligned}(-3 + j5) &= \sqrt{(-3)^2 + (5)^2} \angle \tan^{-1} \left(\frac{5}{-3} \right) \\&= 5.83 + \angle \left[180^\circ - \tan^{-1} \left(\frac{5}{3} \right) \right] \\&= 5.83 + \angle [180^\circ - 59.036^\circ] \\&= 5.83 \angle 120.963^\circ\end{aligned}$$

Step 5 of 5

Substitute $5.83 \angle 120.963^\circ$ for $(-3 + j5)$ in $\frac{(10\angle 60^\circ)(35\angle -50^\circ)}{(2 + j6) - (5 + j)}$.

$$\begin{aligned}\frac{(10\angle 60^\circ)(35\angle -50^\circ)}{(2 + j6) - (5 + j)} &= \frac{350\angle 10^\circ}{5.83 \angle 120.963^\circ} \\&= \left(\frac{350}{5.83} \right) \angle (10^\circ - 120.963^\circ) \\&= 60.03 \angle -110.963^\circ\end{aligned}$$

Therefore, the polar form of $\frac{(10\angle 60^\circ)(35\angle -50^\circ)}{(2 + j6) - (5 + j)}$ is $\boxed{60.03 \angle -110.963^\circ}$.